Let's get into some statistics and probability!

Making Inference and Justifying Conclusions

Here are some terms to familiarize yourself with:
Random Sampling - Usually pertains to a population and is used to get an unbiased and representative sample.
Random sampling can be used to support valid inferences if the sample is representative of the population.

Experimental Probability - Found by repeating an experiment and observing the outcomes.

$$P(\text{event}) = \frac{\text{number of times event occurs}}{\text{total number of trials}}$$

Example:
A coin is tossed 10 times:
A head is recorded 7 times and a tail 3 times.

$$P(\text{head}) = \frac{7}{10}$$

$$P(\text{tail}) = \frac{3}{10}$$

Theoretical Probability - Expresses the likelihood that something favorable will occur.

$$P(\text{event}) = \frac{\text{number of favorable outcomes}}{\text{total number of possible outcomes}}$$

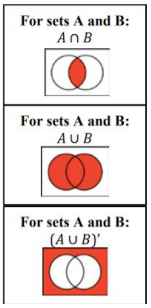
Example:
A coin is tossed.

$$P(\text{head}) = \frac{1}{2}$$

$$P(\text{tail}) = \frac{1}{2}$$

Conditional Probability and the Rules for Probability

Here are some set operations to know:



Intersection - the intersection of two sets A and B is the set of elements that are common to both set A and set B. It is denoted by $A \cap B$ and is read "A intersection B"
Union - the union of two sets A and B is the set of elements, which are in A or in B, or in both. It is denoted by $A \cup B$, and is read "A union B"
Complement - the complement of the set $A \cup B$ is the set of elements that are members of the universal set U but are not in $A \cup B$. It is denoted by $(A \cup B)'$

Conditional Probability - A measure of the probability that an event occurs given that another event has already occurred. **the probability of someone buying car also buying a vacuum

Two-way Tables - A two-way table is a visual representation of the possible relationships between two sets of categorical data. The categories are labeled at the top and the left side of the table, with the frequency (count)

information appearing in the four (or more) interior cells of the table. The "totals" of each row appear at the right, and the "totals" of each column appear at the bottom.

The rows/columns determine the ***condition***. Using the example above, the probability that you select a left-handed person, given that it is a girl is the number of left-handed girls divided by the total number of girls → $P(Left-handed|Girl) = \frac{10}{23} \approx .43$. The ***condition*** in this problem is a ***girl*** therefore, the number of girls represents the total of the conditional probability.

Two-Way Relative Frequency Table

	Girls	Boys	Totals
Left-handed	$\frac{10}{23} \approx .43$	$\frac{15}{27} \approx .56$	$\frac{25}{50} = .50$
Right-handed	$\frac{12}{23} \approx .52$	$\frac{8}{27} \approx .30$	$\frac{20}{50} = .40$
Ambidextrous	$\frac{1}{23} \approx .04$	$\frac{4}{27} \approx .15$	$\frac{5}{50} = .10$
Totals	$\frac{23}{23} = 1.00$	$\frac{27}{27} = 1.00$	$\frac{50}{50} = 1.00$

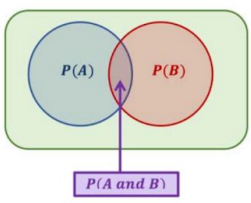
Conditional relative frequencies

Independence - When it is said that two events are independent of each other, it means that the probability of one event occurring does not affect the probability of the other event occurring. Think about it: you rolling a 5 on a dice does not affect you flipping a coin and getting heads.

Joint Probability - The likelihood that two events will occur together.

The Venn diagram shows that when you include everything in both sets the middle region is included twice, therefore you must subtract the intersection region out once. The probability for calculating **joint** events is…

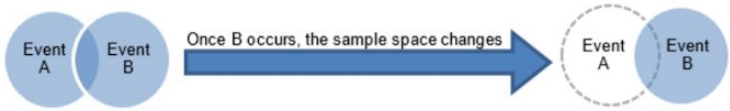
$$P\ (A\ or\ B) = P(A) + P(B) - P\ (A\ and\ B)$$



Students may recognize that if two events A and B are mutually exclusive, also called **disjoint,** the rule can be simplified to $P\ (A\ or\ B) = P\ (A)\ +\ P\ (B)$ since for mutually exclusive events $P\ (A\ and\ B) = 0$.

Marginal Probability - this is the probability of an event without regard to the outcome of another variable.

The sample space of an experiment can be modeled with a Venn diagram such as:



So, the $P(A|B) = \frac{P(A\ and\ B)}{P(B)}$

***Note: Two events must be *independent* to apply the general multiplication rule.**

$$P(A\ and\ B) = P(A)P(B|A) = P(B)P(A|B)$$

The general rule can be explained based on the definitions of independence and dependence. Events are either independent or dependent.

- Two events are said to be **independent** if the occurrence of one event does not affect the probability of the occurrence of the other event.
- Two events are **dependent** if the occurrence of one event does, in fact, affect the probability of the occurrence of the other event.

Making Inference and Justifying Conclusions

- A **population** consists of everything or everyone being studied in an inference procedure. It is rare to be able to perform a census of every individual member of the population. Due to constraints of resources it is nearly impossible to perform a measurement on every subject in a population.

- A **random sample** is a sample composed of selecting from the population using a chance mechanism. Often referred to as a simple random sample.

- **Inferential statistics** considers a subset of the population. This subset is called a statistical sample often including members of a population selected in a random process. The measurements of the individuals in the sample tell us about corresponding measurements in the population.

Observational study - a researcher collects information about a population by measuring a variable of interest but does not impose a treatment on the subjects. (i.e. examining the health effects of smoking).

Example: A department store manager wants to know which of two advertisements is more effective in increasing sales among people who have a credit card with the store. A sample of 100 people will be selected from the 5,300 people who have a credit card with the store. Each person in the sample will be called and read one of the two advertisements. It will then be determined if the credit card holder makes a purchase at the department store within two weeks of receiving the call.

a) Describe the method you would use to determine which credit card holders should be included in the sample. Provide enough detail so that someone else would be able to carry out your method.
b) For each person in the sample, the department store manager will flip a coin. If it lands heads up, advertisement A will be read. If it lands tails up, advertisement B will be read. Why would the manager use this method to decide which advertisement is read to each person?

Source: https://locus.statisticseducation.org/

Experiment - an investigator imposes a change or treatments on one or more group(s), often called treatment group(s). A comparative experiment is where a control group is given a placebo to compare the reaction(s) between the treatment group(s) and the control group.

Margin of Error - how many percentage points your results will differ from the real population value. **taking larger population samples decrease the margin of error
Example question: 1000 people were surveyed and 380 thought that climate change was not caused by human pollution. Find the MoE for a 90% confidence interval.

Step 1: **Find P-hat** by dividing the number of people who responded positively. "Positively" in this sense doesn't mean that they gave a "Yes" answer; It means that they answered according to the statement in the question. In this case, 380/1000 people (38%) responded positively.

Step 2: **Find the z-score that goes with the given confidence interval.** You'll need to reference this chart of common critical values. A 90% confidence interval has a z-score (a critical value) of 1.645.

Step 3: **Insert the values into the formula and solve:**

$$z * \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$= 1.645 * \sqrt{\frac{.38(.62)}{1000}}$$

= 1.645 * 0.0153

= 0.0252

Step 4: **Turn Step 3 into a percentage**:

0.0252 = 2.52%

The margin of error is 2.52%.

Formula:

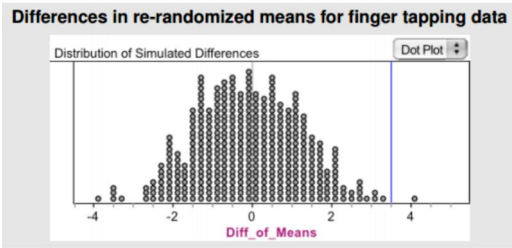
$\hat{p}$ = sample proportion ("P-hat"),
n = sample size,
z = z-score.

$$z * \sqrt{\frac{\hat{p}(1-\hat{p})}{n}},$$

Parameter of Interest - The data that you are focused on in contrast to an entire population. **you can determine the differences in two populations' parameter of interest by finding observable differences in graphs and creating a distribution.

Example: Consider the experiment where twenty male students were randomly assigned to one of two treatment groups of 10 students each, one group receiving 200 milligrams of caffeine and the other group no caffeine. The parameter of interest is the number of finger taps per minute. The sample statistics showed that

the mean of the 200 mg group was 3.5 taps more than the 0 mg group. Thus, an observed difference. Use simulation to determine if the observed difference is due to the caffeine. Is it possible that the 3.5 taps was due to randomization and not caffeine? In order to find out, re-randomize the participants and calculate the difference in means. Simulate this and create a distribution of the results.



The results of the simulation shows that the difference of 3.5 is equaled or exceeded only once out of 400 trials providing strong evidence that the caffeine is the cause of the increased tapping.